

STAAR 2.0 READY

STAAR Science Task Cards

Grade 8

32 TEKS-Aligned Practice Questions

- ★ STAAR 2.0 format practice
- ★ Visual aids & diagrams
- ★ Answer key included
- ★ Recording sheet

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TEKS 8.5A

An element has an atomic number of 12 and a mass number of 24. How many protons, neutrons, and electrons does a neutral atom of this element have?

- A) 12 protons, 12 neutrons, 12 electrons B) 12 protons, 24 neutrons, 12 electrons
C) 24 protons, 12 neutrons, 24 electrons D) 12 protons, 12 neutrons, 24 electrons

2

TEKS 8.5A

Use the portion of the periodic table shown. Which two elements would have the MOST similar chemical properties?

Group 1	Group 2	Group 13	Group 14	Group 15	Group 16	Group 17	Group 18
H (1)							He (2)
Li (3)	Be (4)	B (5)	C (6)	N (7)	O (8)	F (9)	Ne (10)
Na (11)	Mg (12)	Al (13)	Si (14)	P (15)	S (16)	Cl (17)	Ar (18)

- A) Lithium (Li) and Beryllium (Be) B) Lithium (Li) and Sodium (Na)
C) Carbon (C) and Nitrogen (N) D) Helium (He) and Lithium (Li)

3

TEKS 8.5B

In the chemical equation $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$, what do the coefficients represent?

- A) The number of atoms in each molecule B) The number of molecules involved in the reaction
C) The atomic mass of each element D) The speed of the reaction

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TEKS 8.5C

A student performs four different tests in the lab. Which observation provides evidence that a chemical change has occurred?

- A) Ice melting into water B) Sugar dissolving in tea
C) A nail rusting when left outside D) Cutting a piece of paper in half

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TEKS 8.5C

Students tested four substances and recorded their observations. Which substance MOST LIKELY underwent a chemical change?

Substance	Before Heating	After Heating	Observations
A	Solid ice cube	Liquid water	No color change, no gas produced
B	Solid butter	Liquid butter	No color change, no odor
C	White powder	Black solid	Gas bubbles produced, new odor
D	Rock salt	Rock salt crystals	No change observed

- A) Substance A B) Substance B
C) Substance C D) Substance D

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TEKS 8.6A

A 2 kg ball is kicked with a force of 10 N. What is the acceleration of the ball? (Use $F = ma$)



- A) 2 m/s^2 B) 5 m/s^2
C) 10 m/s^2 D) 20 m/s^2

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TEKS 8.6A

A box is being pushed to the right with a force of 50 N. Friction pushes back with a force of 30 N. What is the net force and direction of the box's acceleration?



- A) 80 N to the right B) 20 N to the right
C) 20 N to the left D) 30 N to the left

8

TEKS 8.6A

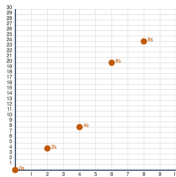
According to Newton's Third Law, when a swimmer pushes water backward with her hands, what happens?

- A) The water pushes the swimmer forward with an equal force B) The water pushes the swimmer backward
C) Nothing happens to the swimmer D) The water disappears

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TEKS 8.6B

A student records the position of a toy car every 2 seconds. Based on the graph, during which time interval was the car moving the fastest?



- A) 0 to 2 seconds B) 2 to 4 seconds
C) 4 to 6 seconds D) 6 to 8 seconds

10

TEKS 8.6B

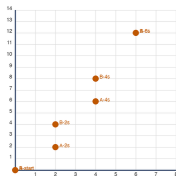
A car travels 150 kilometers in 3 hours. What is the average speed of the car?

- A) 30 km/h B) 45 km/h
C) 50 km/h D) 450 km/h

11

TEKS 8.6B

The graph shows the motion of two runners during a race. Which statement is BEST supported by the data?



- A) Runner A had a constant speed the entire race B) Runner B was faster than Runner A at all times
C) Runner A accelerated during the race while Runner B had constant speed D) Both runners had the same average speed

12

TEKS 8.6C

A wave has a frequency of 5 Hz and a wavelength of 3 meters. What is the speed of the wave? (Use $v = f \times \lambda$)

- A) 2 m/s B) 8 m/s
C) 15 m/s D) 1.67 m/s

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TEKS 8.6C

Students measured the wavelength and frequency of four different waves. Which wave has the GREATEST speed?

Wave	Frequency (Hz)	Wavelength (m)	Speed (m/s)
Wave 1	10	2	20
Wave 2	5	6	30
Wave 3	8	4	32
Wave 4	12	3	36

- A) Wave 1 B) Wave 2
C) Wave 3 D) Wave 4

14

TEKS 8.7A

Which layer of Earth is responsible for the movement of tectonic plates?

- A) Inner core B) Outer core
C) Mantle (asthenosphere) D) Crust

15

TEKS 8.7A

The map shows the location of earthquakes and volcanoes along a boundary. What type of plate boundary is MOST LIKELY shown?

Feature	Observation
Earthquakes	Deep, frequent, along a line
Volcanoes	Chain of volcanic mountains parallel to coast
Ocean Floor	Deep ocean trench present
Rock Age	Older rocks on land, younger near trench

- A) Divergent boundary B) Convergent boundary
C) Transform boundary D) No plate boundary exists here

16

TEKS 8.7B

Which event would MOST LIKELY occur at a divergent plate boundary?

- A) A deep ocean trench forms B) Mountains are pushed upward
C) New ocean floor is created at a mid-ocean ridge D) One plate slides under another

17

TEKS 8.7C

Which of the following correctly lists objects in our universe from SMALLEST to LARGEST?

- A) Galaxy → Solar System → Star → Planet B) Planet → Star → Solar System → Galaxy
C) Star → Planet → Galaxy → Solar System D) Solar System → Planet → Star → Galaxy

18

TEKS 8.7C

A student collects data on the planets in our solar system. Based on the data, which planet would have the STRONGEST gravitational pull on an object on its surface?

Planet	Mass (relative to Earth)	Diameter (km)	Surface Gravity (m/s ²)
Mercury	0.06	4,879	3.7
Earth	1.00	12,742	9.8
Mars	0.11	6,779	3.7
Jupiter	317.8	139,820	24.8

- A) Mercury B) Earth
C) Jupiter D) Mars

19

TEKS 8.8A

Which cell structure is found in plant cells but NOT in animal cells?

- A) Cell membrane B) Nucleus
C) Cell wall D) Mitochondria

20

TEKS 8.8A

A student observes cells under a microscope and records the following data. Which sample is MOST LIKELY from a plant?

Sample	Cell Wall	Chloroplasts	Cell Membrane	Nucleus
Sample 1	No	No	Yes	Yes
Sample 2	Yes	Yes	Yes	Yes
Sample 3	No	No	Yes	No
Sample 4	Yes	No	Yes	No

- A) Sample 1 B) Sample 2
C) Sample 3 D) Sample 4

21

TEKS 8.8B

In humans, brown eye color (B) is dominant over blue eye color (b). A father with genotype Bb and a mother with genotype Bb have a child. What is the probability that the child will have blue eyes?

- A) 0% B) 25%
C) 50% D) 75%

22

TEKS 8.8B

A Punnett square is used to predict offspring traits. Two parents who are both heterozygous (Rr) for round seed shape are crossed. The data table shows the expected results. What percentage of offspring are expected to have round seeds?

	R	r
R	RR	Rr
r	Rr	rr

- A) 25% B) 50%
C) 75% D) 100%

23

TEKS 8.8C

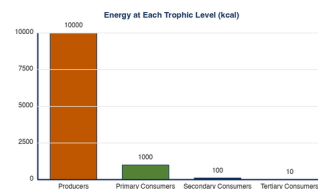
In an ecosystem, grass captures energy from the sun. Grasshoppers eat the grass. Frogs eat the grasshoppers. Snakes eat the frogs. Which organism in this food chain has the LEAST amount of available energy?

- A) Grass B) Grasshopper
C) Frog D) Snake

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TEKS 8.8C

The bar graph shows the amount of energy available at each trophic level in an ecosystem. Approximately what percentage of energy is transferred from producers to primary consumers?



- A) 1% B) 10%
C) 50% D) 90%

25**TEKS 8.8D**

A population of beetles lives on a dark forest floor. Some beetles are dark-colored and some are light-colored. Birds eat the beetles they can see. Over many generations, what will MOST LIKELY happen?

- A) All beetles will become light-colored B) The population will have more dark-colored beetles
C) The birds will stop eating beetles D) The beetles will grow larger to avoid predators

26**TEKS 8.8D**

Scientists tracked a beetle population over 50 years. The data shows the percentage of dark-colored beetles over time. Which concept BEST explains the trend shown?

Year	% Dark Beetles	% Light Beetles	Predator Population
2000	30%	70%	Stable
2010	45%	55%	Stable
2020	60%	40%	Stable
2030	72%	28%	Stable
2050	85%	15%	Stable

- A) Genetic engineering B) Selective breeding
C) Natural selection D) Artificial selection

27**TEKS 8.9A**

Which factor is MOST responsible for the differences in climate between a city near the equator and a city near the North Pole?

- A) Altitude B) Latitude
C) Longitude D) Distance from a lake

28**TEKS 8.9A**

Students collected average temperature and precipitation data for four cities. Which city MOST LIKELY has a desert climate?

City	Avg. Temp (°F)	Annual Rainfall (in)	Latitude
City A	55	45	48°N
City B	78	60	5°N
City C	32	20	65°N
City D	95	4	25°N

- A) City A B) City B
C) City C D) City D

29**TEKS 8.9B**

A student tracks the movement of a cold front on a weather map. Which weather conditions would the student MOST LIKELY observe as the cold front passes through their area?

- A) Clear skies and rising temperatures B) Heavy rain, thunderstorms, then cooler temperatures
C) Fog and steady drizzle for several days D) No change in weather conditions

30**TEKS 8.9C**

A factory releases large amounts of carbon dioxide into the atmosphere. Which long-term effect is this MOST LIKELY to contribute to?

- A) Stronger magnetic field B) Increase in average global temperatures
C) Decrease in ocean levels D) More earthquakes

31**TEKS 8.9C**

The graph shows carbon dioxide levels and average global temperature over 60 years. What relationship is BEST supported by the data?

Year	CO ₂ (ppm)	Avg. Global Temp (°C)
1960	317	13.9
1980	339	14.1
2000	370	14.4
2010	390	14.6
2020	414	14.8

- A)** As CO₂ increases, temperature decreases
- B)** CO₂ levels and temperature are not related
- C)** As CO₂ increases, average global temperature also increases
- D)** Temperature controls CO₂ levels

32**TEKS 8.6A**

A student places a 5 kg object on a frictionless surface and applies different forces. The data is shown below. Based on Newton's Second Law, what force is needed to accelerate the object at 6 m/s²?

Force (N)	Mass (kg)	Acceleration (m/s ²)
5	5	1
10	5	2
15	5	3
20	5	4
?	5	6

- A)** 11 N
- B)** 25 N
- C)** 30 N
- D)** 35 N

Answer Key

#	TEKS	Answer	Correct Choice
1	TEKS 8.5A	A	A) 12 protons, 12 neutrons, 12 electrons
2	TEKS 8.5A	B	B) Lithium (Li) and Sodium (Na)
3	TEKS 8.5B	B	B) The number of molecules involved in the reaction
4	TEKS 8.5C	C	C) A nail rusting when left outside
5	TEKS 8.5C	C	C) Substance C
6	TEKS 8.6A	B	B) 5 m/s ²
7	TEKS 8.6A	B	B) 20 N to the right
8	TEKS 8.6A	A	A) The water pushes the swimmer forward with an equal force
9	TEKS 8.6B	C	C) 4 to 6 seconds
10	TEKS 8.6B	C	C) 50 km/h
11	TEKS 8.6B	C	C) Runner A accelerated during the race while Runner B had constant speed
12	TEKS 8.6C	C	C) 15 m/s
13	TEKS 8.6C	D	D) Wave 4
14	TEKS 8.7A	C	C) Mantle (asthenosphere)
15	TEKS 8.7A	B	B) Convergent boundary
16	TEKS 8.7B	C	C) New ocean floor is created at a mid-ocean ridge
17	TEKS 8.7C	B	B) Planet → Star → Solar System → Galaxy
18	TEKS 8.7C	C	C) Jupiter
19	TEKS 8.8A	C	C) Cell wall
20	TEKS 8.8A	B	B) Sample 2
21	TEKS 8.8B	B	B) 25%
22	TEKS 8.8B	C	C) 75%
23	TEKS 8.8C	D	D) Snake
24	TEKS 8.8C	B	B) 10%
25	TEKS 8.8D	B	B) The population will have more dark-colored beetles
26	TEKS 8.8D	C	C) Natural selection
27	TEKS 8.9A	B	B) Latitude
28	TEKS 8.9A	D	D) City D
29	TEKS 8.9B	B	B) Heavy rain, thunderstorms, then cooler temperatures
30	TEKS 8.9C	B	B) Increase in average global temperatures

#	TEKS	Answer	Correct Choice
31	TEKS 8.9C	C	C) As CO ₂ increases, average global temperature also increases
32	TEKS 8.6A	C	C) 30 N

Student Recording Sheet

Name: _____ Date: _____

#1 A B C D	#2 A B C D	#3 A B C D	#4 A B C D
#5 A B C D	#6 A B C D	#7 A B C D	#8 A B C D
#9 A B C D	#10 A B C D	#11 A B C D	#12 A B C D
#13 A B C D	#14 A B C D	#15 A B C D	#16 A B C D
#17 A B C D	#18 A B C D	#19 A B C D	#20 A B C D
#21 A B C D	#22 A B C D	#23 A B C D	#24 A B C D
#25 A B C D	#26 A B C D	#27 A B C D	#28 A B C D
#29 A B C D	#30 A B C D	#31 A B C D	#32 A B C D